Run report: join-q7pre-run-20261024T2239Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part JOIN under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: join-q7pre 20261024T2239Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 efce43f3fdaa848389a03f3d778a9b9604fe23d1d6e5e1bd75e777a49fb6cf01
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 65702f045ec07ff04edc651d7add4725ceecc7cf1bdc79b495a712a8cb942a44
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/join-q7pre-run-20261024T2239Z JOIN 7 /tmp/claude-0/g4/empty_mass.csv research/genus4_n4_census_20261024/results/tian-q7-
run-20261024T2239Z/tian_by_t.csv
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 0
```

tool_log.txt

```
q=7: prime-order traces with an elliptic curve: 20 (ordinary 20); traces with no hit: [-75, -71, -65, -57, -45, -39, -15, -9, 3, 9, 13, 19, 25, 31, 45, 51, 55, 61, 69, 93]
Spearman(mass, H) = NaN
Spearman(mass, Haar density) = NaN
Spearman(mass, Haar*H/sqrt(disc)) = NaN
Spearman(mass, #l with Tian kernel (depends = 1/2)) = NaN
fit mass = c*H: c = 0.000000e0, R^2 (through origin, ratio-of-sums c) = NaN
fit mass = c*haar: c = 0.000000e0, R^2 (through origin, ratio-of-sums c) = NaN
fit mass = c*haar*H/sqrt: c = 0.000000e0, R^2 (through origin, ratio-of-sums c) = NaN
```

join_summary.txt

```
q=7: prime-order traces with an elliptic curve: 20 (ordinary 20); traces with no hit: [-75, -71, -65, -57, -45, -39, -15, -9, 3, 9, 13, 19, 25, 31, 45, 51, 55, 61, 69, 93]
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fit mass = c*H: c = 0.000000e0, R^2 (through origin, ratio-of-sums c) = NaN
fit mass = c*haar: c = 0.000000e0, R^2 (through origin, ratio-of-sums c) = NaN
fit mass = c*haar*H/sqrt: c = 0.000000e0, R^2 (through origin, ratio-of-sums c) = NaN
```

join_by_t.csv (first 12 of 20 rows)

t	nE	ordinary	hits	mass	mass_se	H	haar_density	arith_x_haar	tian_yes	tian_depends	tian_no
−75	2477	true	0	0.0000	0.0000	20.000	0.048655	0.015427	2	0	1
−71	2473	true	0	0.0000	0.0000	17.333	0.055147	0.014151	3	0	0
−65	2467	true	0	0.0000	0.0000	24.000	0.063845	0.020892	2	0	1
−57	2459	true	0	0.0000	0.0000	16.000	0.073555	0.014763	2	0	1
−45	2447	true	0	0.0000	0.0000	16.000	0.084258	0.015485	2	0	1
−39	2441	true	0	0.0000	0.0000	16.000	0.087955	0.015653	2	0	1
−15	2417	true	0	0.0000	0.0000	24.000	0.092688	0.022970	2	0	1
−9	2411	true	0	0.0000	0.0000	12.000	0.091579	0.011261	2	0	1
3	2399	true	0	0.0000	0.0000	16.000	0.086943	0.014201	2	0	1
9	2393	true	0	0.0000	0.0000	12.000	0.083522	0.010271	3	0	0
13	2389	true	0	0.0000	0.0000	32.000	0.080869	0.026642	2	0	1
19	2383	true	0	0.0000	0.0000	20.000	0.076376	0.015888	3	0	0

Graph

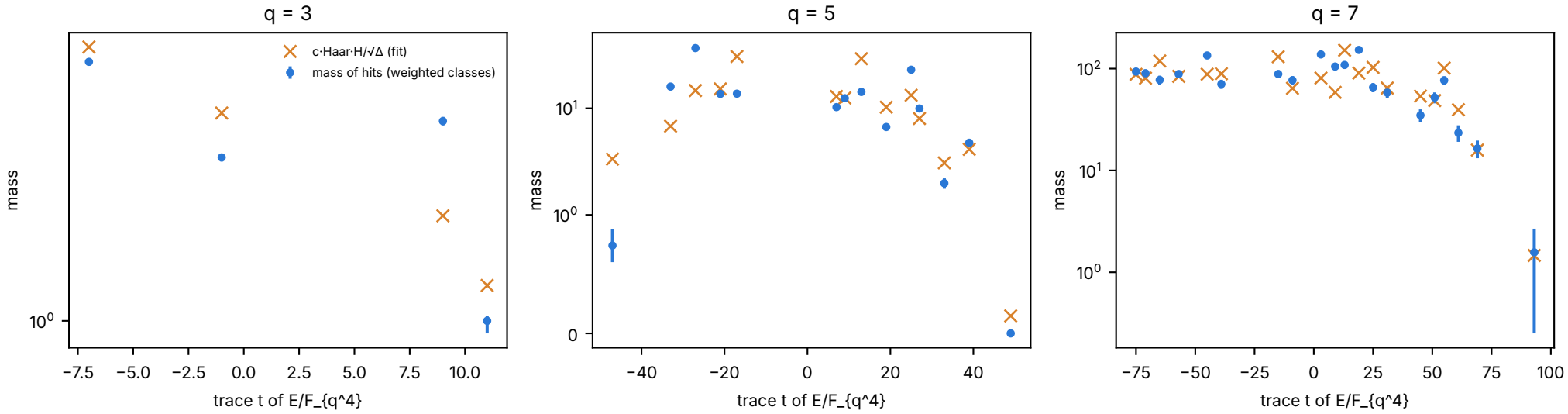


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.